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Spatial and temporal effects on human health following a serious natural disaster --Manuscript Draft--

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Abstract:	Background: Although short-term effects of a disaster on patient health and medical services in a geographically limited area have been previously reported, studies investigating long-term effects on patient health and medical services after a disaster in a large area are scarce. We analyzed the long-term geospatial and temporal effects of The Great East Japan Earthquake. Methods: A retrospective cross-sectional analysis of the Japanese Diagnosis Procedure Combination (DPC) database from August 1, 2010, to March 31, 2015, was done. We conducted a pre-post analysis of the number of patients hospitalized in the DPC-participating hospitals and geospatial analysis using Poisson regression. Findings: The number of inpatients in DPC-participating hospitals showed an apparent decrease with the incidence rate ratio of 0.46 (25th–75th percentile: 0.10–0.80) immediately after the earthquake, and the crude death ratio increased significantly with an odds ratio of 1.78 (25th–75th percentile 1.13–2.26), lasting for 12 months. This change was not only seen in the disaster area but in the broader area of Japan. Interpretation: Owing to their limited capacity, hospitals preferentially treated patients presenting with acute disease, and prolonged medical support reaching a wider area is desirable for future disaster relief measures.

Spatial and temporal effects on human health following a serious natural disaster

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28 **Putting Research into context**

29 **Evidence before this study**

30 We searched PubMed for research on effect of disasters, without language or date
31 restrictions, up to March 31, 2020 using the search terms ((geographical OR spatial)
32 AND temporal natural disaster NOT (climate OR plant OR environment). There was
33 no study investigated the long-term effects of a disaster on patient health or medical
34 services in a large area.

35 **Added value of this study**

36 The present retrospective study analyzed the long-term, nationwide impact on human
37 health following the natural disaster. The crude death ratio increased significantly with
38 an odds ratio of 1.78 (25th–75th percentile 1.13–2.26) not only in the disaster area but
39 in the broader area of Japan. This effect lasted for 12 months after the disaster.

40 **Implications of all the available evidence**

41 The findings of this study suggest that prolonged medical support, reaching a wider
42 area, is desirable for future disaster relief efforts.

43

44 **Abstract**

45 **Background:** Although short-term effects of a disaster on patient health and medical
46 services in a geographically limited area have been previously reported, studies
47 investigating long-term effects on patient health and medical services after a disaster in
48 a large area are scarce. We analyzed the long-term geospatial and temporal effects of
49 The Great East Japan Earthquake.

50 **Methods:** A retrospective cross-sectional analysis of the Japanese Diagnosis Procedure
51 Combination (DPC) database from August 1, 2010, to March 31, 2015, was done. We
52 conducted a pre-post analysis of the number of patients hospitalized in the DPC-
53 participating hospitals and geospatial analysis using Poisson regression.

54 **Findings:** The number of inpatients in DPC-participating hospitals showed an apparent
55 decrease with the incidence rate ratio of 0.46 (25th–75th percentile: 0.10–0.80)
56 immediately after the earthquake, and the crude death ratio increased significantly with
57 an odds ratio of 1.78 (25th–75th percentile 1.13–2.26), lasting for 12 months. This
58 change was not only seen in the disaster area but in the broader area of Japan.

59 **Interpretation:** Owing to their limited capacity, hospitals preferentially treated patients
60 presenting with acute disease, and prolonged medical support reaching a wider area is
61 desirable for future disaster relief measures.

62 **Funding:** The authors received no specific funding for this work.

63

64 **Keywords:** geospatial analysis, temporal modeling, disaster, earthquake, cross-

65 sectional, tsunami, cardiovascular disease

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67

68 **Introduction**

69 Millions of people worldwide are affected by disasters each year. In the past 20 years,
70 natural disasters have claimed the lives of close to 3 million people and have negatively
71 affected the lives of at least 800 million more.¹ The effects of disasters on public health
72 can be classified in the following four categories: (i) direct effects on a population's
73 health, (ii) direct effects on the health care system, (iii) indirect effects on population's
74 health, and (iv) indirect effects on the health care system.²

75 The direct effects of a disaster on population's health result from physical injury and
76 illness. Damage to the health care institution or overload of trauma cases directly affects
77 the health care system. On the contrary, indirect effects on a population's health are
78 related to impaired access to health services due to interruption or overload with or
79 without loss of normal living conditions. The indirect effects on the health care system
80 result from the damage of external infrastructure that the health care system relies upon.

81 Several studies have revealed the influence of natural disasters on disease occurrence,³⁻⁶
82 which involves the worsening of many diseases, especially, the cardiovascular disease.

83 Most of these studies have investigated short period of time after disasters in a
84 geographically limited region, and the results mainly reflect the direct effects of a
85 disaster. Although evidence on the direct effects has been cumulated as the data of

relatively short-term effects on the geographically local area are tractable, studies investigating the indirect effects of a disaster by analyzing large-scale, long-term data are scarce.⁷

Therefore, the present geospatial and long-term temporal analysis aimed to estimate the nationwide, long-term effect of a serious disaster. The study focuses on the changes in the number of patients and their prognosis before and after a disaster, namely, the Great East Japan Earthquake and subsequent tsunamis, which occurred in March 2011 and killed approximately 16,000 people.

Methods

Geographical characteristics and study area

The Great East Japan Earthquake was the most powerful earthquake ever recorded in Japan, and it triggered disastrous tsunamis that reached heights of up to 40.5 m and destroyed vast coastal areas of East Japan. Approximately 16,000 people were killed, and 2,500 people are still missing. The subsequent nuclear accident forced 230,000 people to evacuate their homes into either temporary housing or permanent relocation. A simple outline of the Great East Japan Earthquake is shown in Supplementary Fig S1A.

Japan is divided into 47 prefectures, and each prefecture is further divided into several medical regions (Supplementary Fig S1B and Supplementary Fig S1C, respectively); in total, 344 medical regions exist. The Tohoku region, the main disaster area, is located in the north-eastern part of Japan and is comparably rural with less population (Supplementary Fig S1D). The region comprises six prefectures (Supplementary Fig S1E).

Study design, settings, and data sources

This retrospective cohort study analyzed data of 1,390 hospitals that had participated in a Japanese nationwide administrative database in 2011 after the occurrence of the Great East Japan Earthquake. The participating hospitals were spread across the whole of Japan (Supplementary Fig S1F). The number of patients and crude in-hospital mortality were evaluated. We used the Japanese Diagnosis Procedure Combination (DPC) database, which is a case-mix classification system developed to maintain health policy, including unified payment system, across Japan for acute-phase inpatient hospital care. The DPC-participating hospitals initially comprised 82 hospitals providing advanced treatments,⁸ and the number had increased to account for more than 50% of acute-care hospitals during the study period. The DPC database includes data on all patients

admitted to participating hospitals, including each patient's profile (age, sex, and level of consciousness at the time of admission according to the Japan Coma Scale),⁹ main diagnosis, comorbidities at admission, and post-admission complications, which are independently recorded using relevant codes from the International Classification of Diseases, 10th revision (ICD-10). More details regarding the DPC database have been described elsewhere.^{10,11} Data on region-specific characteristics were collected from the open database of the Ministry of Health, Labour and Welfare.

This study was conducted in accordance with the principles of the 1964 Declaration of Helsinki and its later amendments. The institutional review board of the Tokyo Medical and Dental University approved this study (#788). The requirement for informed consent from each patient was waived because of the retrospective design and the use of anonymized patient data.

Study population

Patients who were admitted to DPC-participating hospitals between August 01, 2010 and March 31, 2015 were included in this study. Of those, we analyzed patients who were admitted to any of the 1,390 hospitals that had participated in the DPC database from the beginning of the observation period.

140

141 *Data collection*

142 The following data were collected from the DPC database: age; sex; ICD-10 codes for
143 four primary diagnoses, concurrent diagnoses at admission, and post-admission
144 complications; unique hospital identifier; and patient survival outcome (survived or
145 deceased). The location of each hospital was identified using Google Maps.¹² The data
146 on population, age distribution, the proportion of smokers, prevalence of obesity,
147 hypertension, and diabetes in each prefecture were collected from the open database of
148 the Ministry of Health, Labour and Welfare.

149

150 *Statistical analysis*

151 We performed pre-post analysis and geospatial analyses. For the former, the number of
152 patients hospitalized per month in each prefecture was analyzed. The Tukey-Kramer
153 method was used to compare the differences between the three time-zones: before the
154 earthquake, within 1 year after the earthquake, and >1 year after the earthquake.

155 For the geospatial analysis, we used Poisson regression considering spatial correlation
156 to reveal spatial pattern behind the data. The model is formulated as

$$y_i \sim \text{Poisson}(\mu_i), \quad \mu_i = \exp\left(\sum_{k=1}^K x_{i,k} \beta_k + z_i\right) \quad (1)$$

$$z_i | z_{j \neq i} \sim N\left(\frac{\sum_j w_{ij}(z_j + \theta_j)}{\sum_j w_{ij}}, \frac{\tau^2}{\sum_j w_{ij}}\right)$$

where the response variable y_i indicates the number of patients and the crude death rate in the i -th medical area. Explanatory variables $x_{i,k}$ ($k \in \{1, \dots, K\}$) were population; age; ratio of patients with obesity, diabetes, and hypertension; and the ratio of smokers in the region. Population was used as the offset variable. β_k represents their coefficients. z_i is a conditional autoregressive (CAR) prior¹³ modeling spatial correlation among neighbors where $w_{ij} = 1$ if medical areas i and j are adjacent, and $w_{ij} = 0$ otherwise. θ_j and τ^2 are parameters estimating similarity with neighbors and the amount of the spatially correlated variations. The model was estimated using the Integrated Nested Laplace approximation,¹⁴ which is a Bayesian estimation approach.

Statistical analyses were performed using R software (version 3.6.0; R Foundation for Statistical Computing, Vienna, Austria). Geospatial analysis was conducted using R-INLA (version 18.07.12).

Role of the funding source

The authors received no specific funding for this work.

Results

Changes in the number of inpatients after the earthquake

Fig 1A illustrates the changes in the number of inpatients in the Tohoku area after the earthquake. The number of inpatients in all prefectures in the area, except the Aomori prefecture, decreased significantly ($p < 0.001$) immediately after the earthquake, lasting for approximately 12 months. The lowest point ranged from 5% to 76% of the average number before the earthquake. This trend was only seen in the Tohoku area; changes in other areas are illustrated in Supplementary Fig S2. Within 1 year after the earthquake, the number of inpatients in the Tohoku area decreased significantly ($p < 0.001$) with the incidence rate ratio of 0.46 (25th–75th percentile: 0.10–0.80) as compared to the period before the earthquake and 1–4 years after the earthquake (Fig 1B).

Changes in inpatient mortality rate after the earthquake

Fig 1C shows the changes in the crude mortality rate in hospitals in the Tohoku area after the earthquake. The crude mortality rate of inpatients increased markedly after the earthquake, and the trend lasted approximately 1 year after the earthquake, which is in line with the period of change in the number of inpatients. The highest point ranged from 1.1- to 4.1-fold increase compared with the crude mortality rate before the

earthquake. The crude mortality rate in the Tohoku area within 1 year after the earthquake increased significantly, with an incidence rate ratio of 1.78 (25th–75th percentile: 1.1–2.3) compared with other time periods ($p < 0.001$) (Fig 1D). The changes in the number of inpatients and crude mortality rate of other areas are shown in Supplementary Fig S2.

Geospatial analysis of each time period

As Fig 2 illustrates, the number of inpatients decreased in the Tohoku area after the earthquake, especially in the Pacific coastal area around the Miyagi prefecture. The mortality rate increased after the earthquake not only in the Tohoku area, but in a wider area including the broad area of north Japan (Fig 3). This result supports the findings shown in Fig 1 and Supplementary Fig S2.

Changes in the distribution of each disease after the earthquake

Fig 4 shows the changes in the proportions of causative diseases (categorized according to ICD-10 codes) for which the patients were hospitalized in the DPC-participating hospitals in the Tohoku area. The results showed that the proportion of cardiovascular disease increased largely within 1 year after the earthquake (13.1% before earthquake,

28.5% after earthquake). There was also a slight increase in the proportion of inpatients admitted for trauma/toxicity (2.2% before earthquake, 4.6% after earthquake) and respiratory disease (13.3% before earthquake, 14.8% after earthquake). Notably, the proportion of inpatients admitted for malignant disease largely decreased (53.8% before earthquake, 34.8% after earthquake).

Changes in the number of inpatients and mortality due to each disease after the earthquake

Fig 5 illustrates the changes in the numbers and crude mortality rates of inpatients, according to the causative diseases, in the Tohoku area after the earthquake. While the number of inpatients decreased, the crude mortality rates increased for all diseases except cerebral hemorrhage. The decrease in the number of inpatients with subarachnoid hemorrhage, acute myocardial infarction, and cerebral hemorrhage was relatively small (incidence rate ratio 0.53, 0.44, and 0.43, respectively), whereas, the crude mortality rates of these diseases remained almost the same as before the earthquake (incidence rate ratio 1.30, 1.89, and 0.66, respectively). On the contrary, the number of inpatients with angina pectoris and old myocardial infarction decreased

markedly (incidence rate ratio 0·08 and 0·03, respectively), and their crude mortality rate apparently increased (incidence rate ratio 9·50 and 12·2, respectively).

Discussion

In this study, we analyzed nationwide administrative data for the long-term, before and after the disastrous earthquake and compared the number of inpatients and crude mortality rate stratified by the diseases. The findings of our study indicated that the impact of disaster on the public health lasted approximately 1 year in relatively broad areas of north Japan. To the best of our knowledge, this is the first study to analyze the long-term effects of a serious natural disaster using geospatial analysis, in terms of patient number and mortality ratio. The data generated in this study would be essential to assess the influence triggered by a serious disaster and could provide important information for further improvement in the disaster management medical system for governments and hospitals.

Although there are a few studies investigating the long-term effects of psychiatric diseases after natural disasters,^{15–17} studies investigating the long-term effects of physical disease after natural disasters are extremely rare. It has been previously reported that a disaster triggers the incidence of cardiological events in the short term.^{18–}

²⁰ Additionally, a study conducted after the Great East Japan Earthquake analyzed ambulance records in the Miyagi prefecture for 4 months after the earthquake. The data revealed a rapid increase, followed by a sharp decrease, in the occurrence of acute coronary syndrome. In contrast, the incidence of heart failure and pneumonia showed a prolonged increase for >6 weeks.⁶ The underlying mechanism remains unclear owing to the limited number of studies investigating this phenomenon. Some reports have claimed that the sympathovagal imbalances,^{21,22} coagulation hyperactivity,²³ or the change of vascular reactivity or microcirculation may be the underlying causes of cardiological events.^{24,25} An observational study of three health care systems 6 months before and 6–16 months after Hurricane Katarina showed that—overall— patients' blood pressure increased after the disaster and gradually returned to baseline levels during the study period, whereas, patients' HbA1c levels increased throughout the observational period.²⁶

In the present study, although the increase in the number of inpatients due to cardiological disease was not observed, the proportion of the patients with cardiological disease increased and the mortality of hospitalized patients increased. This result suggests that hospitals in the disaster area preferentially treated patients with severe acute diseases owing to their limited capacity of receiving patients. Fig 5 represents this

262 interpretation, wherein, the number of inpatients due to life-threatening acute diseases
263 (e.g., subarachnoid hemorrhage, acute myocardial infarction, cerebral hemorrhage) do
264 not show relative decrease, while the number of inpatients with chronic or non-urgent
265 diseases (e.g., angina pectoris, old myocardial infarction) show marked decrease. The
266 decrease in the proportion of malignant diseases in the disaster area also supports this
267 hypothesis.

268 Several direct or indirect factors can affect the capacity of the health care system in the
269 disaster area. After the Great East Japan Earthquake, it was officially announced that at
270 least 11 among the 380 acute-care hospitals in the disaster area completely failed to
271 function, 84 hospitals stopped admitting new inpatients, and 45 hospitals closed their
272 outpatient wards.²⁷ As building damage directly affects the medical capacity, it may be
273 concluded that the hospitals providing tertiary care should be constructed as resistant to
274 mechanical damage. Invisible hazards that hamper the willingness to work among
275 health care providers also result in hospital malfunction.²⁸ After the nuclear accident at
276 Three Mile Island, only six of >70 medical doctors remained to provide service at a
277 hospital.²⁹ After the Great East Japan Earthquake, there was a prominent staff shortage,
278 which lasted for >1 year.³⁰ Due to the aforementioned factors, it is conceivable that
279 DPC-participating hospitals were forced to restrict medical resources for patients with

high priority, whereas, the surrounding non-DPC participating hospitals dedicated their efforts to treat the remaining patients. Although such unavailable patient data may partially affect the results, but we analyzed more than half of the acute-care hospitals, including approximately 95% of tertiary emergency hospitals in Japan; therefore, the results of the present study may be considered to reflect the trends in changes caused by the serious disaster.

This study has several limitations. First, as discussed above, analyzed data were restricted to DPC-participating hospitals. Although the majority of patients with acute and severe diseases were analyzed, the dynamics of non-severe patients were not traceable. Second, although we used population statistics provided by the government, it was unclear whether such data accurately reflect the real population in the disaster area. Finally, as this study assessed the Great East Japan Earthquake, which actually comprised an earthquake, tsunami, and nuclear disaster, generalizability may be considered to be limited. However, to the best of our knowledge, this is the first quantitative estimation of the geographical distribution of effects of a wide-scale disaster. Thus, our data may play an important role in developing future disaster control measures.

To conclude, the Great East Japan Earthquake significantly affected the number of inpatients and mortality rates in the disaster area for at least 1 year, especially for severe acute diseases. Our results suggest that larger scale, prolonged medical support is desirable for future disaster relief efforts. Further complete surveys are required to confirm our results.

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Author Contributions

AS conceived the study, contributed to the design of the study, supervised data collection, did the data analysis, interpreted the results, and wrote the manuscript. AE contributed to the design of the study, supervised data collection, interpreted the results, and revised the manuscript. DM did the data analysis, interpreted the results, and revised the manuscript. HT supported the data analysis and interpreted the results. KF contributed to the design of the study, implemented, supervised data collection, and revised the manuscript. YO contributed to the design of the study, interpreted the

315 results, and revised the manuscript. The corresponding author attests that all listed
316 authors meet the authorship criteria and that no others meeting the criteria have been
317 omitted. AS and YO are the guarantors and had full access to all the data in the study
318 and had final responsibility for the decision to submit for publication.

319

320 **Declaration of interests**

321 The authors declare no conflicts of interests.

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400
401

Figure legends

Figure 1. Changes in the number of inpatients and crude mortality rate of hospitalized patients in Tohoku.

(A) Number of inpatients at each time point of the study. (B) Comparison of the number of inpatients during three time points, before and after the earthquake. (C) Mortality of inpatients plotted at each time point of the study. (D) Comparison of the mortality rate during three time points, before and after the earthquake.

Vertical dashed dotted line indicates the time of the earthquake. Vertical dotted line indicates 12 months after the earthquake.

Figure 2. Geospatial analysis of the changes in the number of inpatients.

(A) 3 months prior to the earthquake (December 2010), (B) 3 months after the earthquake (May 2011), and (C) 2 years after the earthquake (March 2013). Scale bar indicates the incidence rate ratio of the number of inpatients calculated for each region at each time point of the study. The reference was set as the average number of inpatients in each region before the earthquake.

Figure 3. Geospatial analysis of changes in the mortality rate of patients

hospitalized in DPC-participating hospitals.

(A) 3 months prior to the earthquake (December 2010), (B) 3 months after the

earthquake (May 2011), and (C) 2 years after the earthquake (March 2013).

Scale bar indicates the incidence rate ratio of crude death rate calculated for each area at

each time point of the study. The reference was set as the average number of inpatients

in each area before the earthquake.

Figure 4. The proportion of causative diseases leading to hospitalization before and

after the earthquake.

The number of inpatients in Tohoku area was counted after the earthquake, and the

patients were categorized according to the code numbers of ICD-10 (Before earthquake:

1-6 months prior to the earthquake, After earthquake: 1-12 months after the

earthquake).

Figure 5. Changes in the number of inpatients and crude mortality rate of each

disease after earthquake.

436 Number of inpatients and the crude mortality rate were calculated from the inpatients
437 hospitalized within 1 year after the earthquake. The reference point was set as the
438 number and the crude mortality rate before the earthquake. AF: atrial fibrillation, AFL:
439 atrial flutter.

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